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FinTech and small and medium enterprises: current state and future directions

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Abstract

Purpose – Financial technology (FinTech) is the result of the rapid convergence of financial services and advanced technology. FinTech has significantly transformed traditional financial institutions and the financial landscape for small and medium enterprises (SMEs). This study examined the current state and future directions of FinTech adoption and its implications for SMEs.

Design/methodology/approach – Citation analysis, bibliographic coupling and keyword co-occurrence analysis were employed to perform a bibliometric analysis of 125 Scopus-indexed documents published from 2016 to April 2024. The analysis synthesized existing research and identified key themes, influential research and emerging trends in FinTech and SMEs. The study used VOSviewer and Gephi for the analysis and visualization.

Findings – FinTech has introduced novel approaches to credit scoring, alternative financing and digital financial inclusion. However, gaps remain in understanding the full scope of these innovations and their impact on corporate supply chains and Islamic finance. The role of artificial intelligence in revolutionizing credit assessment and the regulatory frameworks governing FinTech companies remains underexplored. The findings highlight the role of FinTech in addressing challenges that SMEs face, such as regulatory complexities and access to financing, thus fostering innovation and economic growth. This study highlights underexplored areas for future research.

Originality/value – The novelty of the study lies in its bibliometric analysis of FinTech-SME literature. It maps the existing body of research to contribute to academic discourse by identifying emerging research trends, gaps and opportunities for future research. The study also provides valuable insights for researchers, policymakers and practitioners on leveraging FinTech to benefit SMEs.

Keywords FinTech, Small- and medium-sized enterprises, Financial inclusion, Crowdfunding, Bibliometric analysis

Paper type Research paper

1. Introduction

In recent years, the convergence of financial services and cutting-edge technology has given rise to a phenomenon known as financial technology, or FinTech (Milian *et al.*, 2019; Pandey *et al.*, 2024). This disruptive force has transformed traditional financial institutions and revolutionized the landscape for small and medium enterprises (SMEs) worldwide (Rehman *et al.*, 2023; Verma *et al.*, 2023). As SMEs are considered the backbone of many economies, understanding the relationship between FinTech and SMEs is imperative for policymakers, financial institutions, and SME owners.

FinTech stems from the fusion of finance and technology, facilitated by rapid advancements in digital infrastructure, data analytics, and artificial intelligence (Belanche *et al.*, 2019). This convergence has led to innovative financial products and services that range from mobile payment systems and peer-to-peer lending platforms to blockchain-based cryptocurrencies and automated wealth management solutions (Kalinic *et al.*, 2019; Laksamana *et al.*, 2022). SMEs, often constrained by limited access to traditional financing channels, stand to benefit significantly from technological innovations (Abbasi *et al.*, 2021a, b; Thomas, 2023).

The integration of FinTech in SMEs is a compelling study area for several reasons. First, SMEs can contribute to economic growth (Surya *et al.*, 2021), foster innovation (Odei and Hamplová, 2022), and create employment opportunities [1]. However, SMEs frequently encounter challenges accessing affordable financing, managing cash flows, and navigating complex regulatory environments (Ghak and Zarrouk, 2022; Jones-Evans, 2015; Osano and Languitone, 2016). Studies suggest that SMEs can leverage FinTech solutions to overcome

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these obstacles, unlocking their full growth potential Abbasi *et al.*, 2021a, b; Sanga and Aziakpono, 2023). With FinTech, even the smallest businesses can access sophisticated financial tools, once accessible only to multinational corporations (Sanga and Aziakpono, 2023). Additionally, it facilitates an ecosystem where entrepreneurs can access funding within minutes, seamlessly conduct cross-border transactions, and analyze real-time financial data to make informed decisions (Saiedi *et al.*, 2022). This vision of a democratized financial landscape powered by FinTech is not merely speculative but a tangible reality, reshaping the SME sector worldwide.

A growing body of literature explores the FinTech and SMEs intersection (Abbasi *et al.*, 2021a, b; Nugraha *et al.*, 2022; Sheng, 2021; Thomas, 2023). Prior studies have investigated the patterns SMEs use to adopt FinTech solutions (Abbasi *et al.*, 2021a, b), how it impacts traditional banking services (Maier, 2016), and the regulatory challenges associated with FinTech innovation. Furthermore, researchers have examined the implications of FinTech for SME financing (Sanga and Aziakpono, 2023), risk management, and business model innovation (Nugraha *et al.*, 2022). While these studies offer valuable insights, and a few studies, have examined the FinTech-SME literature (Sanga and Aziakpono, 2023; Sharma *et al.*, 2024; Verma *et al.*, 2023), owing to the rapid growth in literature, a bibliometric analysis could synthesize existing research and identify emerging trends and research gaps. Hence, this study adopts a bibliometric analysis approach to systematically review the existing literature on FinTech and SMEs. This approach identifies key research themes, influential authors, seminal works, and evolving research trends to offer valuable insights for scholars, practitioners, and policymakers.

There is a need to understand the current state and future directions of FinTech adoption and its implications for SMEs. This study seeks to address this by providing answers to the following questions:

RQ1. Which are the influential documents, journals, authors, and affiliated countries?

RQ2. How has its scholarly discourse evolved?

RQ3. What are the predominant research themes in the intersection of FinTech and SMEs?

RQ4. What are the potential future trajectories of FinTech innovation in the SME sector?

The novelty of the study lies in its bibliometric analysis of the FinTech-SME literature; although undertaken by some studies (Sanga and Aziakpono, 2023; Sharma *et al.*, 2024; Verma *et al.*, 2023), considerably more exploration is needed. Table A1 compares the significant contributions of this study with those of previous studies. The findings have practical implications for policymakers, financial institutions, and SME stakeholders, as they offer valuable insights into harnessing FinTech to benefit SMEs and foster inclusive economic growth.

The rest of the paper presents the data and methods (Section 2), findings of the descriptive analysis (Section 3) and network analysis (Section 4), insights from the content analysis (Section 5), future research agendas (Section 6), implications (Section 7) and the conclusion (Section 8).

2. Data and methods

Researchers employ several literature review methods, including narrative literature reviews to summarize available literature (Sylvester *et al.*, 2013) and systematic literature reviews (SLRs) to answer a set of structured research questions with documents filtered through a pre-defined criterion (Mengist *et al.*, 2020). Given their strength, researchers prefer SLRs to narrative reviews. Most recently, meta-analysis, which extracts statistical data and attempts to validate previous findings with precise estimates, has attracted the attention of researchers

(Pomeroy and Thornton, 2008). This study uses bibliometric analysis (also considered an extension of SLR) (Fan *et al.*, 2022; Lim *et al.*, 2022) to measure the productivity and impact of journals, authors and affiliated countries and to explore the significant themes and collaborating networks in the area of study (Mukherjee *et al.*, 2022). Additionally, it uses quantitative data and statistical analysis to highlight existing and new themes in the domain (Donthu *et al.*, 2021a, b, c; Khan *et al.*, 2022; Mukherjee *et al.*, 2022). Bibliometric analysis is a quantitative technique used to measure publications and the scope of research documents and books (Donthu *et al.*, 2020; Donthu *et al.*, 2021a, b, c). Moreover, bibliometric analysis has been the choice of researchers worldwide when conducting retrospective analyses (Baier-Fuentes *et al.*, 2019; Boakye *et al.*, 2022; Corbet *et al.*, 2019; Flores-Sosa *et al.*, 2022; Merediz-Solá and Bariviera, 2019; Tepe *et al.*, 2022). This approach best fits the goals of this research.

2.1 Tools used for bibliometric analysis

Bibliometric analysis, initially used in library and information sciences (Chang *et al.*, 2015), is a quantitative technique for processing bibliographic data (Broadus, 1987) and using different tools to identify unexplored aspects of a field of study (Pritchard, 1969). While measuring the productivity of a research area (number of publications) (Chung and Cox, 1990; Yazit and Zainab, 2007) or influence (number of citations) (Cancino *et al.*, 2018), it also showcases the worldwide associations between authors with different affiliations. This study uses bibliometric analysis to explore the existing research on FinTech in SMEs. The descriptive analysis includes the total number of documents published, the number of cited publications, and average citations of all published and cited publications. Apart from the descriptive analysis, a citation analysis evaluates the influence of a publication, measured as the number of forward citations received (Gurzki and Woisetschläger, 2017). The study also employs bibliographic coupling (Kessler, 1963), which explains the theme of the studies (Schiuma *et al.*, 2020) to indicate whether all cited and citing documents have similar content. Concomitantly, keyword co-occurrence analysis is conducted to measure the contents of the articles, indicating the conceptual structure of the literature used (Callon *et al.*, 1983; Donthu *et al.*, 2021a, b, c). Apart from these analyses, the most-cited documents, top publishing and cited journals, countries, organizations, and authors are also presented (Albort-Morant and Ribeiro-Soriano, 2016). In network analysis, co-occurrence of author keywords, co-citations, and bibliographic coupling are evaluated using VOSviewer (Van Eck and Waltman, 2017) and Gephi software (Chaurasia and Singh, 2024; Baker *et al.*, 2020; Bastian *et al.*, 2009; Gaur and Kumar, 2018; Khan *et al.*, 2021; Kumari and Jaiswal, 2023; Rai *et al.*, 2024).

2.2 Sample selection and data collection

The data is sourced from the Scopus database for several reasons. First, using the Scopus database avoids selection bias while considering the top-ranked international peer-reviewed journals. Second, Scopus is regarded as the largest multidisciplinary (Baker *et al.*, 2021; Bartol *et al.*, 2014; Kumar *et al.*, 2020; Norris and Oppenheim, 2007) and the most comprehensive citation database (Bar-Ilan, 2010) among all available databases. Third, Scopus is preferred to the Web of Science because it covers "business and economics" (Levine-Clark and Gil, 2008, 2009) and includes many highly cited peer-reviewed journals that are generally not part of the Web of Science (Vieira and Gomes, 2009). Hence, selection bias is minimized. Another more comprehensive database, Google Scholar, provides even more concentrated information (Franceschet, 2010), but the data cannot be processed quickly. Accordingly, following Corbet *et al.* (2019), the Scopus database is used because it offers an exhaustive range of bibliographic data for analysis.

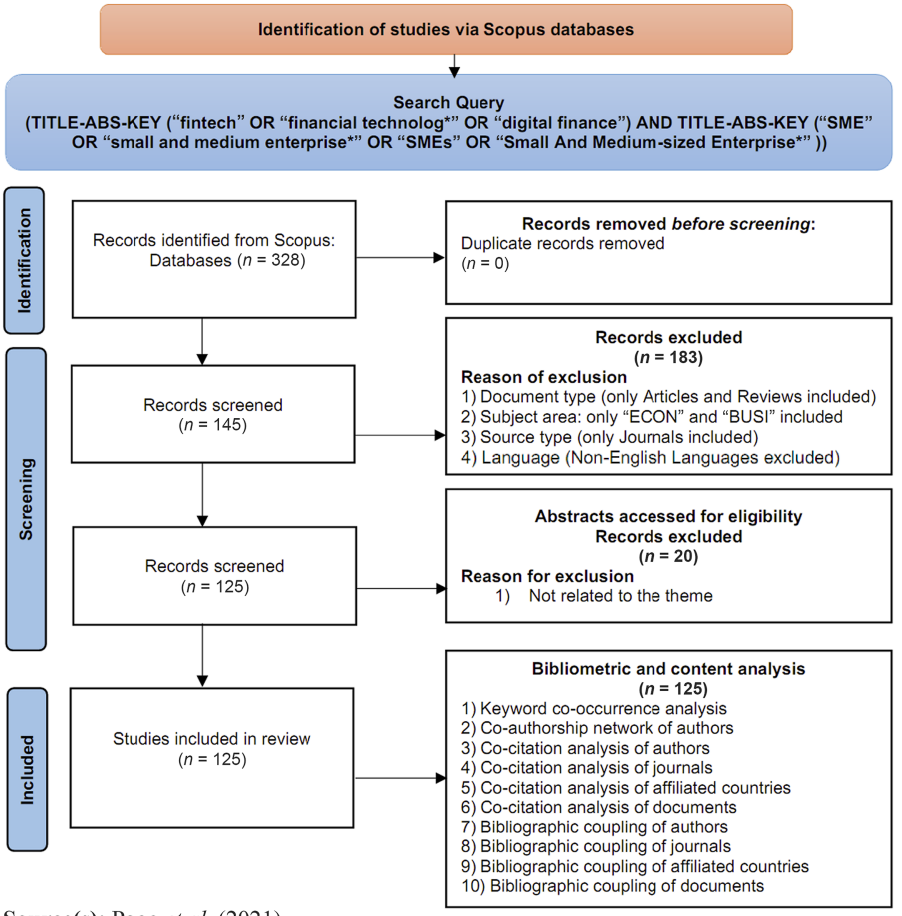
The keywords used for the search include (*"fintech" OR "financial technolog*" OR "digital finance"*) AND (*"SME" OR "small and medium enterprise*" OR "SMEs" OR "Small and Medium-sized Enterprise*"*). The automatic filter is applied to restrict the subject

area to *"Business, Management, and Accounting," "Economics, Econometrics and Finance," and "Social Sciences."* Further, only *"articles"* and *"reviews"* in *"journals"* published in the *"English"* language are retained (Khan *et al.*, 2022; Kumar *et al.*, 2020). The initial search yielded 328 documents; automatic filters reduced this number to 145. Furthermore, content-based exclusion of documents led to 20 more being excluded as they were irrelevant to the study area. Thus, 125 documents were used for bibliometric analysis. Figure 1 presents the sample extraction and exclusion process using the PRISMA framework Page *et al.* (2021).

3. Descriptive analysis

3.1 Publication and citation trends

The FinTech-SME literature has significantly grown from one published document in 2016 to 24 published by April 2024. Since 2016, 125 documents have been published, attracting 2,167 citations, a rate of 17.34 citations per year (Table 1). These findings align with those of prior studies, suggesting that FinTech research growth began in 2016 (Biju *et al.*, 2024; Pandey



Source(s): Page *et al.* (2021)

Figure 1. PRISMA framework

Table 1. Publication and citation trend

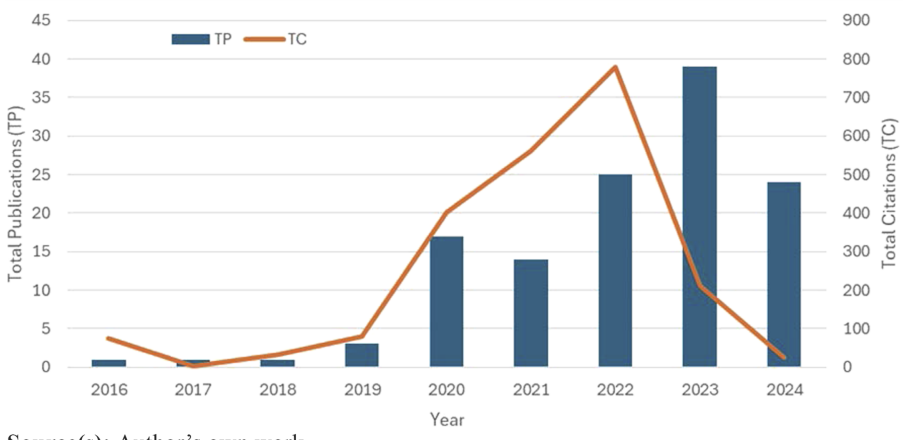
Year	TP	TC	AC
2016	1	75	75.00
2017	1	2	2.00
2018	1	33	33.00
2019	3	80	26.67
2020	17	402	23.65
2021	14	561	40.07
2022	25	779	31.16
2023	39	211	5.41
2024	24	24	1.00
Total	*125*	*2,167*	*17.34*

Note(s): This table presents the year-wise publication and citation trend. TP = total publications, TC = Total Citations, AC = Average citations
Source(s): Author's own work

et al., 2024; Trotta *et al.*, 2024). However, focusing on FinTech-SME literature reveals significant growth from 2019 onward (Figure 2).

3.2 Most-cited documents

An examination of the growth in FinTech-SME literature reveals influential documents. Table 2 presents the top ten cited articles, which span various dimensions and geographical contexts. Akpan *et al*. (2022) is the most-cited article, with 162 citations; the study highlights persistent challenges that SMEs in emerging and developing markets face when adopting advanced technologies, contrasting them with success stories from developed economies. The study emphasizes the role of FinTech and analytical algorithms in enhancing operational efficiencies and competitive advantages crucial for SME sustainability. Pizzi *et al*. (2021), the second most-cited research, with 136 citations, provide empirical insights into how FinTech helps overcome organizational barriers to facilitate smoother transitions and operational improvements. Sheng (2021), the third most-cited study, with 121 citations, focuses on China, demonstrating how FinTech enhances credit access for SMEs, with varying impacts across



Source(s): Author's own work

Figure 2. Publication and citation trend

Table 2. Most cited documents

Authors	Title	Year	Source title	TC
Akpan I.J.; Udoh E.A.P.; Adebisi B.	Small business awareness and adoption of state-of-the-art technologies in emerging and developing markets, and lessons from the COVID-19 pandemic	2022	Journal of Small Business and Entrepreneurship	162
Pizzi S.; Corbo L.; Caputo A.	Fintech and SMEs sustainable business models: Reflections and considerations for a circular economy	2021	Journal of Cleaner Production	136
Sheng T.	The effect of fintech on banks' credit provision to SMEs: Evidence from China	2021	Finance Research Letters	121
Soni G.; Kumar S.; Mahto R.V.; Mangla S.K.; Mittal M.L.; Lim W.M.	A decision-making framework for Industry 4.0 technology implementation: The case of FinTech and sustainable supply chain finance for SMEs	2022	Technological Forecasting and Social Change	100
Hassan M.K.; Rabbani M.R.; Mohd. Ali M.A.	Challenges for the Islamic finance and banking in post COVID era and the role of Fintech	2020	Journal of Economic Cooperation and Development	95
Wu Y.; Huang S.	The effects of digital finance and financial constraint on financial performance: Firm-level evidence from China's new energy enterprises	2022	Energy Economics	92
Lu Z.; Wu J.; Li H.; Nguyen D.K.	Local Bank, Digital Financial Inclusion and SME Financing Constraints: Empirical Evidence from China	2022	Emerging Markets Finance and Trade	90
Abbasi K.; Alam A.; Du M.A.; Huynh T.L.D.	FinTech, SME efficiency and national culture: Evidence from OECD countries	2021	Technological Forecasting and Social Change	83
Maier E.	Supply and demand on crowdlending platforms: connecting small and medium-sized enterprise borrowers and consumer investors	2016	Journal of Retailing and Consumer Services	75
Syed M.H.; Khan S.; Rabbani M.R.; Thalassinos Y.E.	An artificial intelligence and NLP based Islamic FinTech model combining zakat and Qardh-Al-Hasan for countering the adverse impact of COVID 19 on SMEs and individuals	2020	International Journal of Economics and Business Administration	72

Note(s): This table highlights the top cited documents in the FinTech-SME literature
Source(s): Author's own work

bank sizes. Together, the other most-cited studies provide a robust understanding of how FinTech innovations can empower SMEs worldwide, addressing financial inclusion (Lu *et al.*, 2022), operational and financial efficiency (Abbasi *et al.*, 2021a, b; Maier, 2016; Soni *et al.*, 2022; Wu and Huang, 2022), Islamic finance practices (Hassan *et al.*, 2020), and resilience to economic disruptions (Mohammad *et al.*, 2020).

3.3 Productive and influential publication outlets

Publication outlets play a vital role in disseminating research in any domain. Outlets that align with a particular research topic attract more publications. Table 3a presents the top-productive

Table 3. Top productive and influential publication outlets

Source title	TP	TC	AC
(a) *Top productive publication outlets*			
Finance Research Letters	5	125	25.00
Emerging Markets Finance and Trade	5	116	23.20
Investment Management and Financial Innovations	3	39	13.00
Qualitative Research in Financial Markets	3	114	38.00
Technological Forecasting and Social Change	3	188	62.67
Economic Analysis and Policy	3	74	24.67
Journal of Innovation and Knowledge	3	34	11.33
Journal of Open Innovation: Technology, Market, and Complexity	3	70	23.33
International Review of Financial Analysis	3	4	1.33
(b) *Top influential publication outlets*			
Technological Forecasting and Social Change	3	188	62.67
Journal of Small Business and Entrepreneurship	1	162	162.00
Journal of Cleaner Production	1	136	136.00
Finance Research Letters	5	125	25.00
Emerging Markets Finance and Trade	5	116	23.20
Qualitative Research in Financial Markets	3	114	38.00
Journal of Economic Cooperation and Development	1	95	95.00
Energy Economics	1	92	92.00
International Journal of Economics and Business Administration	2	77	38.50
Journal of Retailing and Consumer Services	1	75	75.00

Note(s): These tables provide the top productive and influential journals in the FinTech-SME literature
Source(s): Author's own work

journals that publish at least three documents. *"Finance Research Letters"* and *"Emerging Markets Finance and Trade"* are the most productive journals with five documents each, attracting 125 and 116 citations, respectively. Other journals that have published at least three articles each include *"Investment Management and Financial Innovations," "Qualitative Research in Financial Markets," "Technological Forecasting and Social Change," "Economic Analysis and Policy," "Journal of Innovation and Knowledge," "Journal of Open Innovation: Technology, Market, and Complexity,"* and *"International Review of Financial Analysis."* Higher productivity indicates that the aims and scope of these journals align with those of FinTech-SME literature. Hence, these journals are potential outlets for future research. Table 3b presents the most influential journals in terms of citations. The journals *"Technological Forecasting and Social Change (TC = 188)," "Journal of Small Business and Entrepreneurship (TC = 162)," "Journal of Cleaner Production (TC = 136)," "Finance Research Letters (TC = 125),"* and *"Emerging Markets Finance and Trade (TC = 116)"* are the publication outlets with most citations in FinTech-SME literature. These are followed by *"Qualitative Research in Financial Markets," "Journal of Economic Cooperation and Development," "Energy Economics," "International Journal of Economics and Business Administration,"* and *"Journal of Retailing and Consumer Services."* More citations indicate that publications in these journals are highly acknowledged and referred to by researchers in this domain. Future researchers interested in FinTech-SME research could refer to these outlets for more insights into the field.

3.4 Productive and influential affiliated countries

Affiliated countries play a significant role in research. Countries that expend resources on research and development are considered innovative and resilient to crisis (Kumari *et al.*, 2024). Hence, when countries have numerous studies published in FinTech-SME literature, they may be aligned with the domain. Table 4a presents the list of the most productive affiliated

Table 4. Top productive and influential affiliated countries

Country	TP	TC	AC
(a) *Top productive affiliated countries*			
China	36	667	18.53
Indonesia	26	273	10.50
India	13	178	13.69
Malaysia	11	214	19.45
United Kingdom	11	596	54.18
United States	11	486	44.18
Italy	6	216	36.00
Australia	5	148	29.60
Bahrain	4	177	44.25
Ghana	4	13	3.25
South Africa	4	21	5.25
(b) *Top influential affiliated countries*			
China	36	667	18.53
United Kingdom	11	596	54.18
United States	11	486	44.18
Indonesia	26	273	10.50
Italy	6	216	36.00
Malaysia	11	214	19.45
India	13	178	13.69
Bahrain	4	177	44.25
Germany	3	176	58.67
Nigeria	3	173	57.67
Viet Nam	3	173	57.67

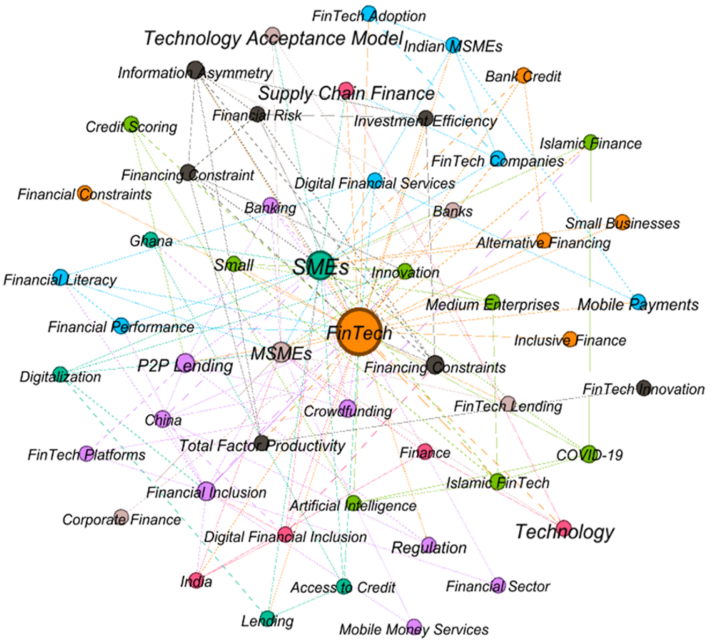
Note(s): These tables present the top productive (number of documents) and influential (number of citations) affiliated countries in the FinTech-SME literature
Source(s): Author's own work

countries. The top three most productive affiliated countries—China (TP = 36), Indonesia (TP = 26), and India (TP = 13)—have 667, 273, and 178 citations, respectively. These are followed by Malaysia, the United Kingdom, and the United States, each with 11 documents. Findings suggest that emerging nations have become the most productive countries in FinTech-SME literature. Table 4b presents the most influential affiliated countries with the most citations. China (TC = 667), the United Kingdom (TC = 596), the United States (TC = 486), Indonesia (TC = 273), and Italy (TC = 216) are the top five most influential affiliated countries.

4. Network analysis

4.1 Keyword co-occurrence analysis

The keyword co-occurrence analysis creates a network of frequently occurring author keywords to segregate the themes of the extant literature (Liu and Mei, 2016; Maltseva and Batagelj, 2020). Documents with common keywords form separate clusters, which indicate the thematic alignment of the literature. Figure 3 presents the co-occurrence network of 52 author keywords that appear in at least two documents. The keywords *"FinTech-83," "SMEs-36," "MSMEs-15," "financial inclusion-11," "P2P lending-11," "financial constraints-10," "crowdfunding-8," "information asymmetry-8," "COVID-19–7,"* and *"China-6"* are the most frequently occurring author keywords, indicating the thematic evolution of the FinTech-SME literature. This suggests that the existing literature focuses on the intersection of FinTech with SMEs and MSMEs, highlighting the roles of financial inclusion, peer-to-peer (P2P) lending, and crowdfunding in mitigating financial constraints and information asymmetry, including in the context of COVID-19 and its impact on China.



Note(s): Keyword co-occurrence – 373 keywords, at least 2 occurrences – 52, all connected

Source(s): Author's own work (generated using VOSviewer and Gephi)

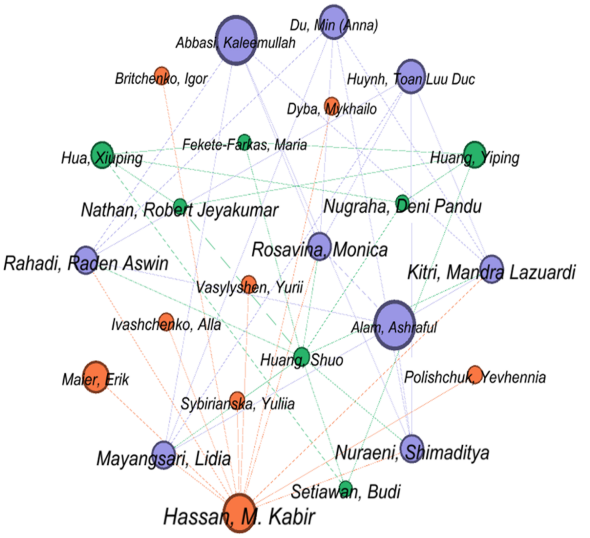
Figure 3. Keyword co-occurrence network

4.2 Citation analysis

Citation analysis evaluates the relationship between scholarly documents, authors, sources, and affiliated countries (Zupic and Čater, 2015). Citation analysis helps understand the structure and dynamics of scholarly communication networks, identify important works and research trends, and gain insights into the impact and influence of individual documents or authors within a research field. This study performs citation analyses of authors, sources, affiliated countries, and documents.

4.2.1 Authors' citation analysis. Figure 4 illustrates the citation network of 24 out of 87 authors with at least 20 citations. The top authors in terms of citations are *Abbasi, K. (TC = 129), Alam, A. (TC = 129), Hassan, M.K. (TC = 98), Du, M.(A.) (TC = 83), Huynh, T.L.D. (TC = 83)*, and *Maier, E. (TC = 75). Hassan, M.K. (WL = 12), Huang, S. (WL = 9), Kitri, M.L. (WL = 6), Mayangsari, L. (WL = 6), Nuraeni, S. (WL = 6), Rahadi, R.A. (WL = 6),* and *Rosavina, M. (WL = 6)* have more weight links (WL) [2] in literature. *Abbasi, K.* and *Alam, A.* published two documents examining how FinTech, particularly peer-to-peer (P2P) lending, impacts the efficiency of and access to finance for SMEs (Abbasi *et al.*, 2021a, b), while also considering the moderating effects of national culture and institutional quality (Abbasi *et al.*, 2021a, b). *Hassan, M.K.* published two documents with a focus on integrating FinTech with Islamic finance and SMEs, particularly addressing the challenges posed by the COVID-19 pandemic (Hassan *et al.*, 2020) and advancing future research agendas to foster sustainable financial systems and regulatory frameworks (Verma *et al.*, 2023).

4.2.2 Journals' citation analysis. Figure 5 shows the citation network of 36 connected sources from among the 93 potential sources based on total link strength (TLS) [3]. The journals *"Economic Analysis and Policy (TLS = 10)," "Emerging Markets Finance and*



Note(s): Citation analysis of authors – 1 document 20 citations, 87 met, 24 connected

Source(s): Author's own work (generated using VOSviewer and Gephi)

Figure 4. Citation analysis of authors

Note(s): Citation analysis of journals – 1 document 0 citations, 93 met, 36 connected

Source(s): Author's own work (generated using VOSviewer and Gephi)

Figure 5. Citation analysis of journals

Trade (TLS = 9)," *"Qualitative Research in Financial Markets (TLS = 9),"* *"Digital Business (TLS = 8),"* and *"European Management Journal (TLS = 7)"* emerge as highly connected journals This indicates that these journals significantly influence research in the FinTech-SME literature, serving as key nodes in the citation network.

4.2.3 Affiliated countries' citation analysis. Figure 6 presents the citation network of 31 affiliated countries from 40 countries. *"China (TLS = 46),"* *"India (TLS = 37),"* *"United States (TLS = 26),"* *"Indonesia (TLS = 22),"* *"South Africa (TLS = 16),"* *"Australia (TLS = 15),"* *"Malaysia (TLS = 12),"* and *"United Kingdom (TLS = 12)"* are highly connected with other countries. Highly connected countries tend to produce research with broader impact, as their work is more likely to be cited by international peers. This reinforces the importance of fostering global research networks to enhance the visibility and influence of scholarly work.

4.2.4 Documents' citation analysis. Figure 7 displays the citation network of 44 connected documents from among the 125 analyzed. The documents Rosavina *et al*. (2019) and Huang (2022) with WLs of nine each, Sanga and Aziakpono (2023) with a WL of 8, Lu *et al*. (2022) and Verma *et al*. (2023) with a WL of seven each, Maier (2016) with a WL 6, and Thomas (2023) with a WL of 5, are among the top linked documents in the FinTech-SME literature. Highly linked documents serve as pillars, influencing and shaping ongoing studies. The frequent citation of these articles highlights their importance in providing a theoretical basis, empirical evidence, or methodological advancements critical for further research.

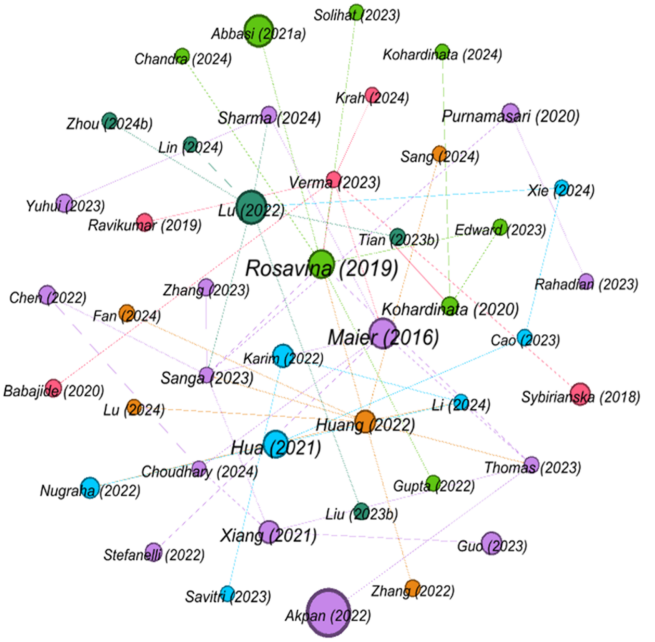
4.3 Bibliographic coupling

Bibliographic coupling refers to the degree of intellectual similarity or thematic relatedness between two documents, where similarity is based on shared references (Jarneving, 2007; Ni

Note(s): Citation analysis of countries – 1 document 0 citation, 40 met, 31 connected

Source(s): Author's own work (generated using VOSviewer and Gephi)

Figure 6. Citation analysis of countries



Note(s): Citation analysis of documents – 0 citations, 125 met, 44 connected

Source(s): Author's own work (generated using VOSviewer and Gephi)

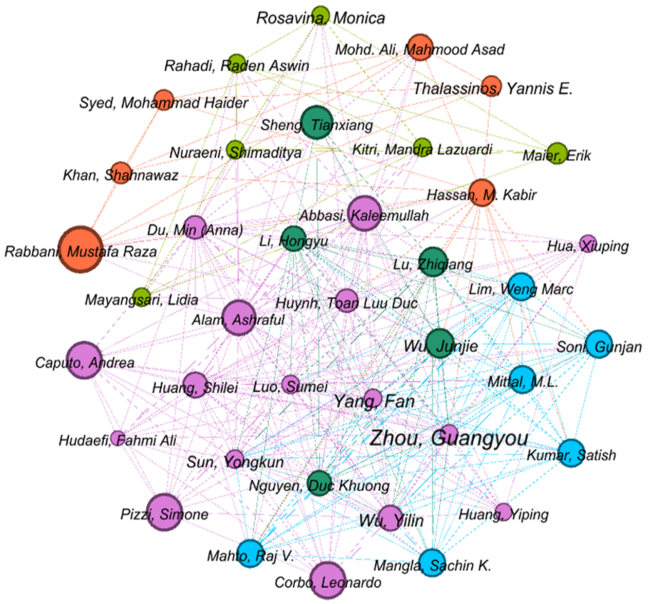
Figure 7. Citation analysis of documents

et al., 2013). In other words, bibliographic coupling occurs when two documents cite the same or similar sources. This study uses bibliographic coupling of authors, sources, affiliated countries, and documents.

4.3.1 Authors' bibliographic coupling. Figure 8 illustrates the bibliographic network of 39 connected authors from among 42 authors with at least 50 citations. *Abbasi, K., Alam, A., Du, M.(A.), Huynh, T.L.D., Kumar, S., Lim, W.M., Mahto, R.V., Mangla, S.K., Mittal, M.L., Soni, G.,* and *Rabbani, M.R.,* are highly associated authors in terms of common referencing in the FinTech-SME literature. These authors share numerous common references, indicating their intellectual similarity. Future researchers may consider these connections for collaborative research.

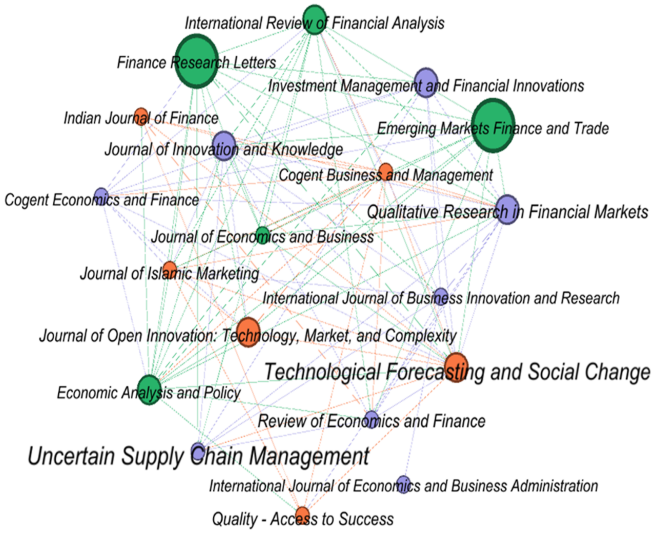
4.3.2 Journals' bibliographic coupling. Figure 9 shows the bibliographic network of 19 connected sources with at least two documents. *"Emerging Markets Finance and Trade," "International Review of Financial Analysis," "Economic Analysis and Policy," "Technological Forecasting and Social Change," "Qualitative Research in Financial Markets,"* and *"Finance Research Letters,"* have high TLS indicating that they are closely connected in terms of the references in the documents they publish. Future researchers can target these journals for their FinTech-SME domain research output.

4.3.3 Affiliated countries' bibliographic coupling. Figure 10 presents the bibliographic network of 39 affiliated countries out of the 40 countries. *China, the United States, India, Malaysia, the United Kingdom,* and *Australia* are highly connected in terms of intellectual contributions. Authors from these countries share a common intellectual network as they cite common papers in their research. A mix of developed and emerging nations share common intellectual networks in FinTech-SME literature.



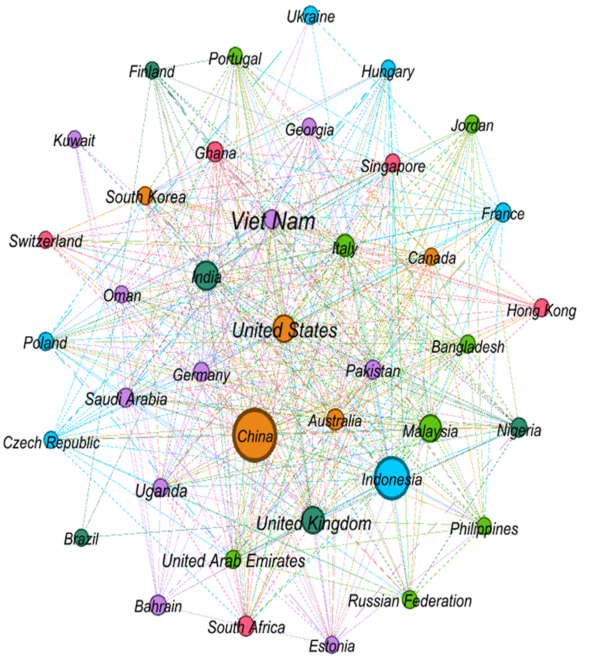
Note(s): Bibliographic coupling of authors-1 document 50 citation, 42 met, 39 connected
Source(s): Author's own work (generated using VOSviewer and Gephi)

Figure 8. Bibliographic coupling of authors



Note(s): Bibliographic coupling of journals – Total 93 sources, 2 documents 0 citations, 19 met, 19 connected
Source(s): Author's own work (generated using VOSviewer and Gephi)

Figure 9. Bibliographic coupling of journals



Note(s): Bibliographic coupling of countries – Total 40, 1 document 0 citations, 40 met, 39 connected

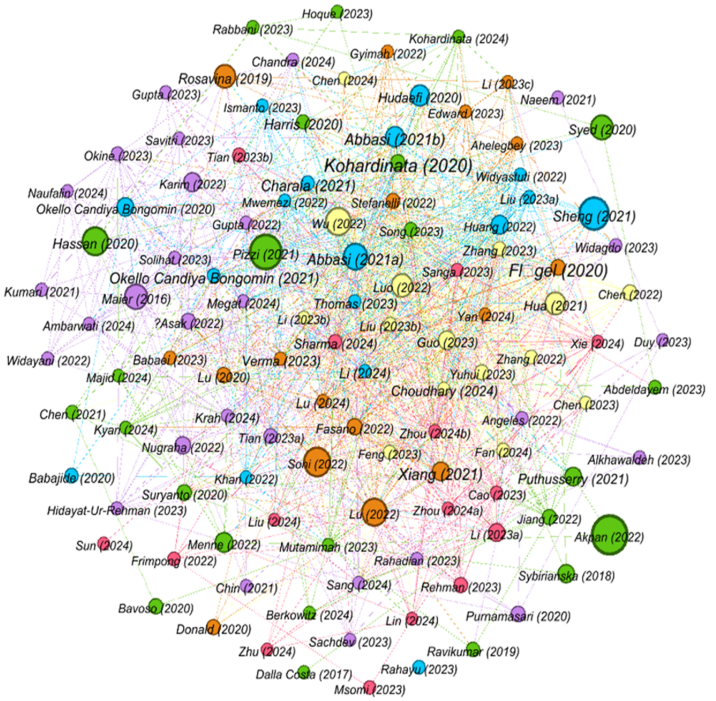
Source(s): Author's own work (generated using VOSviewer and Gephi)

Figure 10. Bibliographic coupling of countries

4.3.4 Documents' bibliographic coupling. Figure 11 displays the bibliographic coupling of 116 connected documents from 125 documents. These documents are classified into six thematic clusters, indicating their special focus on different aspects of the FinTech-SME literature. The documents by Gupta *et al.* (2022), Song *et al.* (2023), Huang (2022), Y. Lu *et al.* (2024), Guo *et al.* (2023), and Sharma *et al.* (2024) have the highest TLS among their clusters. Gupta *et al.* (2022) examine the role of FinTech in the financial inclusion of Indian MSMEs, finding high acceptance is influenced by prior experience, brand familiarity, government support, and behavioral variables, while demographic factors were ineffective. Song *et al.* (2023) reveal the role of FinTech in supply chain finance as a signaling intermediary that reduces information asymmetry. Huang (2022) finds that FinTech improves investment efficiency in Chinese SMEs, especially in standardized information-processing regions. Lu *et al.* (2024) highlight the dual effects of FinTech on corporate investment sensitivity to interest rates in China, with market-driven reforms needed for efficacy. Guo *et al.* (2023) reveal that FinTech reduces financing constraints in Chinese SMEs through enhanced financing and competition. Sharma *et al.* (2024) review the impact of FinTech on small businesses, identifying usage and impact themes and suggesting further research areas. Insights from these studies could help future researchers direct their research.

5. Content analysis

The bibliographic coupling of the 116 connected documents allows dividing them into six clusters; these studies provide the basis for classifying the current state of FinTech-SME literature. The most-cited papers from each of the six clusters are selected for content analysis to provide deeper insights into the thematic focus of the clusters. Future research questions are



Note(s): Bibliographic coupling of documents – Total 125, 0 citations, 125 met, 116 connected. At least 10 papers in a cluster, forms 6 clusters. Purple, Green, Blue, Orange, Yellow, Pink

Source(s): Author's own work (generated using VOSviewer and Gephi)

Figure 11. Bibliographic coupling of documents

suggested based on this analysis. The list of most-cited papers in each cluster is available in Table A2.

5.1 Cluster 1. FinTech adoption and impact on SMEs and MSMEs

This cluster (in purple) is the largest, with 27 papers. The documents in this cluster highlight the transformative role of FinTech in SME finance. Maier (2016) emphasizes crowdlending platforms as efficient alternatives to traditional banking for SMEs. Karim *et al*. (2022) identify critical factors in ASEAN countries, emphasizing the resilience of FinTech during crises. Nugraha *et al*. (2022) show the drivers of FinTech adoption among Indonesian SMEs, enhancing financial inclusion. Łasak (2022) discusses the broader implications of FinTech for SMEs in developing countries, emphasizing efficiency and service expansion. Purnamasari *et al*. (2020) explore FinTech adoption barriers in Indonesia, focusing on education and business size challenges. Although regulatory and educational barriers persist, FinTech emerges as pivotal in democratizing SME finance. Further, readiness to adopt innovative systems (such as lean practices) is critical for SMEs (Alnajem *et al*., 2019). Alnajem (2020) emphasizes operational efficiency in banks by adopting lean methodologies in FinTech applications to enhance SME access to financial resources.

5.2 Cluster 2. technological innovations and SMEs in emerging markets

This cluster (in green) contains 24 documents. The studies in the cluster examine the adoption of advanced technologies and financial innovations to bolster SMEs in emerging markets.

Akpan *et al*. (2022) identify barriers to SMEs' adoption of digital technology in developing economies, emphasizing the potential of FinTech in enhancing operational efficiency. Pizzi *et al*. (2021) highlight the role of FinTech in promoting sustainable business models under Industry 4.0, advocating its integration with circular economy practices. Hassan *et al*. (2020) discuss the resilience of Islamic FinTech during crises, proposing its use to support SMEs in Islamic economies post-COVID-19. Syed M.H. *et al*. (2020) propose an artificial intelligence (AI)-driven Islamic FinTech model to mitigate the economic impact of COVID-19 on SMEs. Menne *et al*. (2022) analyze the impact of Sharia-compliant FinTech on SME financial performance in Indonesia.

5.3 Cluster 3. fintech and SME financing: global perspectives and impacts

This cluster (in blue) contains 17 documents. The documents in this cluster highlight the impact of FinTech on small and medium enterprises (SMEs) by enhancing access to finance and operational efficiency. Sheng (2021) finds that FinTech boosts bank credit supply to SMEs in China, particularly benefiting larger banks. Abbasi *et al*. (2021a, b) show that P2P lending FinTechs increase access to finance for SMEs in OECD countries, moderated by institutional quality. Hudaefi (2020) explores Islamic FinTech in Indonesia, promoting sustainable development goals through financial inclusion. Huang (2022) demonstrates the role of FinTech in improving SME investment efficiency in China. These findings underscore the importance of FinTech in supporting SMEs globally, prompting strategic and policy considerations for future development.

5.4 Cluster 4. digital finance innovations for SMEs: global perspectives

This cluster (in orange) contains 17 documents. The studies in this cluster underscore the role of FinTech in enhancing financing and operational efficiency for SMEs. Soni *et al*. (2022) highlight Industry 4.0 technologies like IoT and big data in optimizing supply chain finance, benefiting SMEs. Lu *et al*. (2022) find that digital financial inclusion and local bank branches alleviate financing constraints for SMEs in China. Rosavina *et al*. (2019) explore factors that drive SMEs to adopt P2P lending platforms in Indonesia, emphasizing loan flexibility and alternative payment schemes. Xiang *et al*. (2021) reveal that FinTech adoption varies among Chinese SMEs owing to ownership structures and financial constraints. Fasano and Cappa (2022) discuss how internet banking via FinTech impacts SME debt issuance.

5.5 Cluster 5. digital finance and SME dynamics in China

This cluster (in yellow) contains 15 documents. The studies in this cluster highlight the transformative impact of FinTech on Chinese enterprises, particularly in promoting sustainable development and efficiency gains. Wu and Huang (2022) explore how digital finance enhances financial performance for new energy firms, even though moderated by financial constraints. Luo *et al*. (2022) explore how FinTech innovation boosts total factor productivity across enterprises and emphasize its dual effects on information dissemination and resource allocation, which are more pronounced in small and medium-sized private enterprises. Hua and Huang (2021) identify drivers behind FinTech growth in China, emphasizing its role in enhancing financial inclusion and supporting SMEs. Guo *et al*. (2023) reveal that FinTech significantly alleviates financing constraints, particularly benefiting non-state-owned and innovative firms. Chen *et al*. (2022) use Alibaba data to show that FinTech credit reduces firm volatility and enhances resilience through increased investment during downturns, underlining its pivotal role in the stability and growth of MSMEs.

5.6 Cluster 6. impact of digital finance on SMEs: insights and innovations

This cluster (in pink) contains 15 documents. The studies in this cluster underscore the pivotal role of FinTech in SME financing. Li *et al*. (2023) demonstrate how digital finance (DF) reduces financing constraints for Chinese A-share listed companies, particularly benefiting

SMEs and private enterprises. Rehman *et al.* (2023) highlight the ability of FinTech to enhance bank credit supplies for manufacturing SMEs through blockchain and Big Data technologies. Tian and Shao (2023) find that DF mitigates the financial risks of SMEs in China by improving transparency and efficiency. Sharma *et al.* (2024) summarize key themes from a systematic review, emphasizing the impact of FinTechs on payment innovations and financial services accessibility for small businesses. In Ghana, Frimpong *et al.* (2022) show that DF enhances SME performance by improving financial literacy and access.

6. Future research agendas

The content analysis in Section 5 reveals significant gaps in the literature. The significant role of FinTech in transforming SME finance calls for focused research agendas to address critical gaps and advance the field. FinTech has reshaped traditional banking practices and introduced novel approaches to credit scoring, alternative financing, and digital financial inclusion. Despite these advancements, our understanding of these innovations and their impact on corporate, supply chain, and Islamic finance is insufficient. The role of AI in revolutionizing credit assessment and the regulatory frameworks governing FinTech companies remains underexplored. While service quality in traditional retail sectors like hypermarkets significantly impacts customer satisfaction and loyalty (Elmelegy *et al.*, 2017), a similar emphasis on service quality in the FinTech sector, especially for SMEs adopting digital financial solutions, could be interesting.

Additionally, financial constraints, investment efficiency, and adopting technology acceptance models in mobile payments add complexity to the landscape. Addressing these gaps is crucial for informing policy, enhancing financial sector resilience, and promoting inclusive growth across diverse economies. Investigating how regulatory frameworks impact FinTech adoption among MSMEs in developing countries and understanding the role of financial literacy in overcoming adoption barriers for SMEs in emerging markets is essential. The potential of AI-driven FinTech solutions to support SMEs in Islamic economies during economic crises presents a promising research avenue. Moreover, examining the factors that contribute to the successful integration of FinTech with sustainable business models in emerging markets is vital. Finally, understanding how the shift from soft to complex information impacts SME debt issuance through internet banking and exploring how FinTech innovation affects the productivity and financial resilience of private versus state-owned SMEs during economic downturns are essential areas for future research. These are transformed into the future research questions (FRQs) below.

FRQ1. How do regulatory frameworks impact FinTech adoption among MSMEs in developing countries?

FRQ2. What roles do financial inclusion and literacy play in overcoming FinTech adoption barriers for SMEs in emerging markets?

FRQ3. How can AI-driven FinTech solutions support SMEs in Islamic economies during economic crises?

FRQ4. What factors drive successful integration of FinTech with sustainable business models in emerging markets?

FRQ5. How does the shift from soft to hard information impact SME debt issuance through internet banking?

FRQ6. How does FinTech innovation affect private versus state-owned SME productivity and financial resilience during economic downturns?

FRQ7. How do FinTech lending practices differ from traditional bank credit offerings in promoting inclusive finance?

These FRQs aim to guide future researchers in exploring under-researched areas at the intersection of banking, finance, innovation, and technology, mainly focusing on their implications for SMEs and emerging markets.

7. Implications of the study

The findings of this study offer significant theoretical and practical implications. From a theoretical perspective, the study contributes to the growing FinTech-SME literature by elucidating the transformative role of FinTech in addressing SME-specific challenges such as financial constraints (Wu and Huang, 2022; Xiang *et al.*, 2021) and information asymmetry (Tian and Shao, 2023). These insights align with and extend the existing literature on financial inclusion (Hudaefi, 2020; Hua and Huang, 2021) and innovation diffusion in SMEs (Sharma *et al*., 2024), emphasizing the potential of digital solutions like P2P lending (Abbasi *et al.*, 2021a, b; Rosavina *et al.*, 2019), crowdfunding, and credit access (Sheng, 2021). Further, the differential focus on FinTech research in emerging and developed markets underscores the need for further research into contextual factors, such as regional regulatory frameworks (Jones-Evans, 2015; Purnamasari *et al.*, 2020; Verma *et al.*, 2023). Furthermore, the cluster analysis aided with the content analysis maps the existing FinTech-SME literature, making it more accessible for researchers to carry forward the theoretical discourse through the suggested FRQs.

From the practical perspective, for FinTech leaders, the findings suggest that the operational and financial challenges of SMEs in emerging markets could be addressed by prioritizing innovation (Luo *et al.*, 2022; Odei and Hamplová, 2022). SMEs in developing regions must develop strategies to overcome barriers to adopting advanced technologies (Hu *et al.*, 2024). Policymakers can facilitate this by adopting supportive ecosystems through streamlined regulations, infrastructure development, and targeted financial literacy programs (Frimpong *et al.*, 2022; Verma *et al.*, 2023). These measures are essential to ensure SME sustainability, resilience, and competitive advantage in the face of global disruptions and rapidly evolving market dynamics.

8. Conclusion

This study offers a bibliometric analysis of the intersection between FinTech and SMEs, highlighting the transformative impact of FinTech on the SME sector. The analysis reveals that FinTech innovations provide significant opportunities for SMEs to overcome financial constraints and enhance their operational efficiencies. The existing literature strongly focuses on financial inclusion, peer-to-peer lending, and crowdfunding. This study identifies key influential documents, journals, and countries contributing to this domain, underscoring the importance of global collaboration in advancing FinTech research. The findings suggest that although the existing research has progressed substantially, gaps and emerging trends call for further investigation. Future research should continue to explore the nuanced impacts of FinTech on SMEs across different regions and economic contexts, as well as the evolving regulatory challenges. Policymakers and financial institutions should consider these insights to create supportive environments that facilitate SMEs' adoption of FinTech solutions, ultimately contributing to inclusive economic growth. Understanding and leveraging the benefits of FinTech can help SMEs achieve greater financial resilience and drive sustainable economic development. Supplementary file presents a summary of the findings.

Notes

1. https://smestreet.in/msme-opportunities/smes-drive-innovation-and-employment-in-export-markets-6943670
2. WL is the number of connections that an author or document possesses.
3. TLS is the sum of the link strengths of a node over other nodes.

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Appendix 1

Table A1. A comparison with available reviews

Particulars	Sharma *et al*. (2024)	Verma *et al*. (2023)	Sanga and Aziakpono (2023)	Present study
Title	Small businesses and FinTech: a systematic review and future directions	FinTech in small and medium enterprises (SMEs): A review and future research agenda	FinTech and SMEs financing: A systematic literature review and bibliometric analysis	FinTech and small and medium enterprises: current state and future directions
Authors	Sharma, Sachin Kumar; Ilavarasan, P. Vigneswara; Karanasios, Stan	Verma, Sushma; Shome, Samik; Hassan, M. Kabir	Sanga, Bahati; Aziakpono, Meshach	–
Source	Electronic Commerce Research	European Management Journal	Digital Business	–
Sample period	2008-Early 2023	1988–2021 (November)	2008–2022	2016–2024 (May)
Database	Scopus, Web of Science, EBSCOhost, Science direct, SAGE, Willey-Blackwell, Taylor & Francis, and Springer databases	Scopus and Web of Science	Scopus and World of Science	Scopus
Keywords used for document search	"small businesses," "small enterprises," "small firms," "SME," "microbusinesses," micro-business, "microenterprises," "micro-enterprises", "micro-firms", "small entrepreneurs", "microentrepreneurs", "micro-entrepreneurs", "retailers, and "merchants" and "FinTech", "mobile payment", "digital payment", "mobile money", "mobile wallet", "digital lending", "digital loan", "online lending", "online loan", "digital finance", "online finance", "internet-based finance", "peer to peer lending", "P2P lending", "peer to peer loan", "P2P loan", "crowdfunding", "crowdlending", "cryptocurrency", "crypto payment", "initial coin offering", "insurtech", "regtech"	TITLE-ABS-KEY ("FinTech" OR "FinTechs" OR "fin-tech" OR "fin-techs" OR "financial technology" OR "financial technologies" OR "finance technology" OR "finance technologies" OR "Financial Innovation" OR " Digital Finance" AND "SME" OR "Small Business" OR "Small and Medium Enterprises" OR "SMEs" OR "MSMEs" OR "Micro, Small and Medium Enterprises" OR "Micro Business" OR "Micro Enterprises" OR "Small Enterprises" OR "Micro Business")	"Financial Technology" OR "Blockchain" OR "Mobile" OR "Digital" OR "Crowdfunding" OR "Crowd-lending" OR "Crowd-investing" OR "Marketplace Lending" OR "Peer-to-Peer lending" OR "P2P" OR "Alternative Financing" AND "SMEs" OR "SME" OR "Small Businesses" OR "Small Firms" OR "Small and Medium Enterprises" AND "Lending" OR "Borrowing" OR "Finance" OR "Credit" OR "Loan" OR "Trade Credit" OR "Bank Lending" OR "Bank Financing" OR "Bank Loan" OR "Bank Credit" OR "Capital" OR "Venture Capital" OR "Debt" OR "Equity" OR "Debt Financing" OR "Equity Financing" OR "Overdraft Facility"	TITLE-ABS-KEY ("fintech" OR "financial technolog*" OR "digital finance") AND TITLE-ABS-KEY ("SME" OR "small and medium enterprise*" OR "SMEs" OR "Small And Medium-sized Enterprise*")
Sample size	103	108	62	125
Methodology	Systematic review	Bibliometric review	Bibliometric review	Bibliometric review and content analysis
Source included	Only journal articles in 4*, 4, 3, and 2 ranked journals in the Academic Journal Guide (AJG) 2021 list	Only peer-reviewed articles	Only peer-reviewed articles	Only peer-reviewed articles and reviews
Date published	May-23	Aug-23	Oct-23	–
Focus	Small businesses	SMEs	SME Financing	SMEs

Source(s): Author's own work

Appendix 2

Table A2. Documents in different clusters

Authors	Title	Source title	TC
Cluster 1. FinTech adoption and impact on SMEs and MSMEs			
Maier (2016)	Supply and demand on crowdlending platforms: connecting small and medium-sized enterprise borrowers and consumer investors	Journal of Retailing and Consumer Services	75
Karim *et al*. (2022)	Is FinTech providing effective solutions to Small and Medium Enterprises (SMEs) in ASEAN countries?	Economic Analysis and Policy	38
Nugraha *et al*. (2022)	Fintech Adoption Drivers for Innovation for SMEs in Indonesia	Journal of Open Innovation: Technology, Market, and Complexity	27
Łasak (2022)	The role of financial technology and entrepreneurial finance practices in funding small and medium-sized enterprises	Journal of Entrepreneurship, Management and Innovation	17
Purnamasari *et al*. (2020)	Technology Acceptance Model of Financial Technology in Micro, Small, and Medium Enterprises (MSME) in Indonesia	Journal of Asian Finance, Economics and Business	17
Cluster 2. Technological innovations and SMEs in emerging markets			
Akpan *et al*. (2022)	Small business awareness and adoption of state-of-the-art technologies in emerging and developing markets, and lessons from the COVID-19 pandemic	Journal of Small Business and Entrepreneurship	162
Pizzi *et al*. (2021)	Fintech and SMEs sustainable business models: Reflections and considerations for a circular economy	Journal of Cleaner Production	136
Hassan *et al*. (2020)	Challenges for the islamic finance and banking in post COVID era and the role of Fintech	Journal of Economic Cooperation and Development	95
Syed *et al*. (2020)	An artificial intelligence and NLP based Islamic FinTech model combining zakat and Qardh-Al-Hasan for countering the adverse impact of COVID 19 on SMEs and individuals	International Journal of Economics and Business Administration	72
Menne *et al*. (2022)	Optimizing the Financial Performance of SMEs Based on Sharia Economy: Perspective of Economic Business Sustainability and Open Innovation	Journal of Open Innovation: Technology, Market, and Complexity	42
Cluster 3. Fintech and SME financing: global perspectives and impacts			
Sheng (2021)	The effect of fintech on banks' credit provision to SMEs: Evidence from China	Finance Research Letters	121
Abbasi *et al*. (2021a, b)	FinTech, SME efficiency and national culture: Evidence from OECD countries	Technological Forecasting and Social Change	83
Hudaefi (2020)	How does Islamic fintech promote the SDGs? Qualitative evidence from Indonesia	Qualitative Research in Financial Markets	50
Abbasi *et al*. (2021a, b)	P2P lending Fintechs and SMEs' access to finance	Economics Letters	46
Huang (2022)	Does FinTech improve the investment efficiency of enterprises? Evidence from China's small and medium-sized enterprises	Economic Analysis and Policy	36

(*continued*)

Table A2. Continued

Authors	Title	Source title	TC
Cluster 4. Digital finance innovations for SMEs: global perspectives			
Soni *et al*. (2022)	A decision-making framework for Industry 4.0 technology implementation: The case of FinTech and sustainable supply chain finance for SMEs	Technological Forecasting and Social Change	100
Lu *et al*. (2022)	Local Bank, Digital Financial Inclusion and SME Financing Constraints: Empirical Evidence from China	Emerging Markets Finance and Trade	90
Rosavina *et al*. (2019)	P2P lending adoption by SMEs in Indonesia	Qualitative Research in Financial Markets	64
Xiang *et al*. (2021)	Determinants of the use of fintech finance among chinese small and medium-sized enterprises	IEEE Transactions on Engineering Management	36
Fasano and Cappa (2022)	How do banking fintech services affect SME debt?	Journal of Economics and Business	22
Cluster 5. Digital finance and SME dynamics in China			
Wu and Huang (2022)	The effects of digital finance and financial constraint on financial performance: Firm-level evidence from China's new energy enterprises	Energy Economics	92
Luo *et al*. (2022)	Does fintech innovation promote enterprise transformation? Evidence from China	Technology in Society	64
Hua and Huang (2021)	Understanding China's fintech sector: development, impacts and risks	European Journal of Finance	60
Guo *et al*. (2023)	FinTech and financing constraints of enterprises: Evidence from China	Journal of International Financial Markets, Institutions and Money	36
Chen *et al*. (2022)	Finance and Firm Volatility: Evidence from Small Business Lending in China	Management Science	17
Cluster 6. Impact of digital finance on SMEs: insights and innovations			
Li *et al*. (2023)	Digital finance and enterprise financing constraints: Structural characteristics and mechanism identification	Journal of Business Research	31
Rehman *et al*. (2023)	FinTech Adoption in SMEs and Bank Credit Supplies: A Study on Manufacturing SMEs	Economies	14
Tian and Shao (2023)	Financing Constraints and Information Asymmetry of SMEs — the Development of Digital Finance and Financial Risks of Enterprises	Journal of the Knowledge Economy	8
Sharma *et al*. (2024)	Small businesses and FinTech: a systematic review and future directions	Electronic Commerce Research	8
Frimpong *et al*. (2022)	Financial literacy, access to digital finance and performance of SMEs: Evidence From Central region of Ghana	Cogent Economics and Finance	6

Source(s): Author's own work

Supplementary material

The Supplementary material for this article can be found online.

About the author

Dr Dharen Kumar Pandey is an Assistant Professor at the Indian Institute of Management Sambalpur, Odisha. With over 12 years of industry and academic experience, he has previously served as an Inspector of GST, worked with the Income Tax Department, Government of India, and as an Assistant Professor in Magadh University, Bodh Gaya, Bihar. Dr Pandey earned his Ph.D. from Banaras Hindu University, and Post-Doctoral from Faculty of Business, Economics, and Social Development, University Malaysia Terengganu. His research interests include financial markets and econometrics, with publications in top-tier journals including the *Economic Letters*, *International Review of Economics and Finance*, *Journal of Behavioral and Experimental Finance*, *Financial Research Letters*, *Research in International Business and Finance*, *Journal of Business Research*, among others. His four research papers have been awarded in the International Research Awards hosted by the Institute of Chartered Accountants of India. Dharen Kumar Pandey can be contacted at: dharenp@gmail.com

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